Auditable Monocular Human–Object Reconstruction with Evidence-Constrained Repairs

Extended Abstract

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We reconstruct human geometry, object meshes and continuous object motion from 30 monocular V2D demonstrations using pretrained visual models, optimization and targeted repairs. Across thirteen submitted variants, the retained T1-11 candidate reduces object Chamfer distance from 25.60394 to 24.21106 cm and object acceleration error from 1.74271 to 0.49967 cm/frame^2 relative to our initial submission. Human geometry improves slightly, while penetration worsens. We contribute an auditable sequence of positive and negative experiments, source-constrained repair mechanisms and reusable interfaces. The results support cautious geometric/temporal correction, not universal improvement from additional optimization or a different pretrained body estimator.

1. Introduction

Monocular human–object reconstruction must resolve identity, geometry, scale, occlusion and motion from ambiguous observations. V2D Track 1 [2] evaluates both reconstruction accuracy and physical plausibility against withheld multi-view estimates. The output comprises MHR human parameters, object geometry and per-frame object poses, exported on the required scored frame grid. Track 1 and Track 2 episode indices describe different sequences; our Track 1 reconstruction does not use Track 2 assets or trajectories.

We build on a CARI4D-style workflow [5] and evaluate targeted changes against an immutable retained baseline. Our contribution is the repair, validation and provenance workflow, rather than a new pretrained vision model. This report designates the complete T1-11 file; later T1-12 and T1-13 trials did not justify replacing it.

2. Approach

Initialization and refinement. The reconstruction stack combines SAM2 segmentation [3], monocular depth, SAM 3D body/object estimates [1], object registration/tracking and joint human–object refinement. We investigated CoCoNet and the released iterative refinement path; later diagnostic pipelines also used MoGe-2 and FoundationPose. Their experiments are distinguished from components that actually produced the retained submission. Human parameters, triangle meshes, object rotations/translations and positive object scale are exported with explicit coordinate conventions and timestamps.

Targeted repairs. We investigated geometry scale/proportion, orientation ambiguity, temporal noise, support consistency and person/object selection. Candidate transforms are bounded by source-image evidence and preserve unmodified intervals. Object-family experiments test shape-appropriate hypotheses instead of applying the same correction to all objects. Accepted changes are layered on the retained baseline with per-episode identity and provenance checks. Occluded frames retain a continuous trajectory as required by evaluation, but inferred motion is not treated as an independently observed pose.

Evaluation discipline. The challenge's first-frame human Sim(3) alignment affects both human and object evaluation. Changing human scale or placement can therefore alter object scores even when object coordinates are unchanged. We compare human, object and physical-plausibility metrics jointly. Visual silhouette agreement, source depth diagnostics and official scores are separate evidence streams. Image-gated repair hypotheses are generated without hidden reference inputs; submission selection uses observed benchmark scores and may overfit this small set. Rejected hypotheses and producing source snapshots remain available.

3. Experiments & Results

We submitted thirteen complete variants spanning initialization/refinement, geometry and temporal corrections, object-family repairs and alternative human

Metric	Initial T1-01	Retained T1-11
CD-H	15.53707	15.47123
CD-O	25.60394	24.21106
ACC-H	1.38088	.27484
ACC-O	1.74271	.49967
PEN	.00430	.00485

Table 1: Kaggle scores of two exact files. The retained file covers all 30 episodes and 9,877 scored frames. These are not independent-seed averages.

estimates. Table 1 gives exact scores of the initial and retained 30-episode files. All metrics are lower-is-better. Chamfer distances and penetration are in centimetres; acceleration discrepancies are in cm/frame^2 under the metric's discrete-frame convention.

The retained result reduces CD-O by 5.44% and ACC-O by 71.33%. CD-H changes little and PEN becomes worse: smoothing and improved pose agreement do not ensure better contacts. An early geometry/temporal variant reduced acceleration error but worsened CD-O to 32.65560 cm; selective rollback recovered object accuracy. Subsequent object-family work decreased CD-O from 24.82940 through 24.31892 to 24.21106 cm. The final retained iron-family change affects Episode 8 while preserving all human values and unselected object values. Later desk/proportion submissions yield CD-O 24.45089 and 24.21206 cm and remain diagnostic alternatives.

Negative studies. More refinement was not uniformly beneficial, motivating bounded comparisons and checkpoint reuse. Alternative human pipelines were tested, including FlowHMR [4] and GVHMR diagnostics. A wrong-person FlowHMR case improved held-out silhouette IoU from 0.3303 to 0.7222 after selection repair, but remained below the retained baseline's 0.8029. Across the all-30 image screen, mean baseline IoU was 0.7855 versus 0.7085 for the corrected official FlowHMR branch. Representation-conversion error, camera-depth ambiguities and body identity must be evaluated separately; a successful conversion check is not ground-truth reconstruction accuracy. These unselected studies are documented but are not claimed as contributors to the retained score.

4. Conclusion

Evidence-constrained, object-specific corrections improve the aggregate reconstruction, but accuracy, temporal fidelity and contact plausibility can conflict. **Economic viability** is addressed through pretrained components, selective recomputation and checkpointed experiments on a laptop/L4 budget; total production cost savings have not been measured. The **data flywheel** stores diagnosed failures, verified repairs and regression cases for reuse, while acknowledging continued manual review. **Complexity reduction** comes from explicit geometry/time/coordinate contracts and modular reconstruction/repair interfaces rather than new capture hardware. **Innovation** is treating repairs as testable hypotheses with rollback and preserving negative results instead of assuming every refinement or model replacement improves reconstruction.

Future work should prioritize metric-depth observability, contact and cross-sequence validation. This is a challenge-scale study, not a demonstrated autonomous production system. Code and provenance: https://github.com/isyed/v2d-reconstruction; reusable modules: https://github.com/isyed/real2sim2real. Exact submission IDs and the producing commit accompany the report; packaged-output verification and raw-video reruns are reported separately.

References

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